

Visualization Tools for Structural Analysis



CS
01

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INTRODUCTION & PROBLEM

- KPFF is a structural engineering firm with more than 64 years of experience.
- KPFF engineers use **OpenSees**, a **command-line based** seismic simulation tool for structural analysis.
- This **lack of visualization** slows workflows.

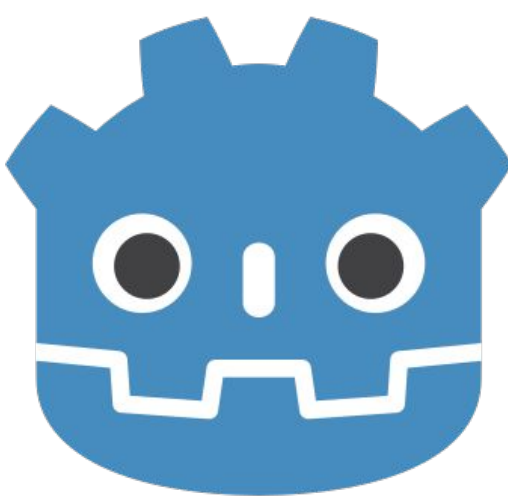
EXISTING SOLUTIONS

- Tools like **RISA-3D®** and **ETABS®** focus on usability, but lack advanced analysis.
- **Perform3D®** is powerful but outdated and unintuitive.
- OpenSees has no GUI, requiring manual text output review and third-party software.

PROJECT GOAL

- Deliver an interactive 3D visualization tool that bridges the gap between OpenSees power and intuitive usability – **all within a single interface**.
- This will enable engineers to visually review models and outputs before and after OpenSees simulations.

TECH STACK



GODOT
Game engine

DEVELOPMENT ROADMAP



Onboarding
Research



Plotly Prototyping
↓
Godot Transition



Godot Framework
GUI Design



3D Rendering
Simulation Features



DCR Visualization
Testing
Deployment

DESIGN IMPLEMENTATION AND FEATURES

```
1 # restraints
2 element elasticBeamColumn 165 1107 1108 165 165
3 element elasticBeamColumn 166 1108 1109 166 166
4 element elasticBeamColumn 167 1109 1110 167 167
5 element elasticBeamColumn 171 1115 1116 171 171
6 element elasticBeamColumn 175 1120 1121 175 175
7 element elasticBeamColumn 176 1121 1122 176 176
8 element elasticBeamColumn 177 1122 1123 177 177
9 element elasticBeamColumn 178 1123 1124 178 178
10 element elasticBeamColumn 179 1124 1125 179 179
11 element elasticBeamColumn 180 1125 1126 180 180
12 element elasticBeamColumn 181 1126 1127 181 181
13 element elasticBeamColumn 182 1127 1128 182 182
14 element elasticBeamColumn 183 1128 1129 183 183
15 element elasticBeamColumn 184 1129 1130 184 184
```

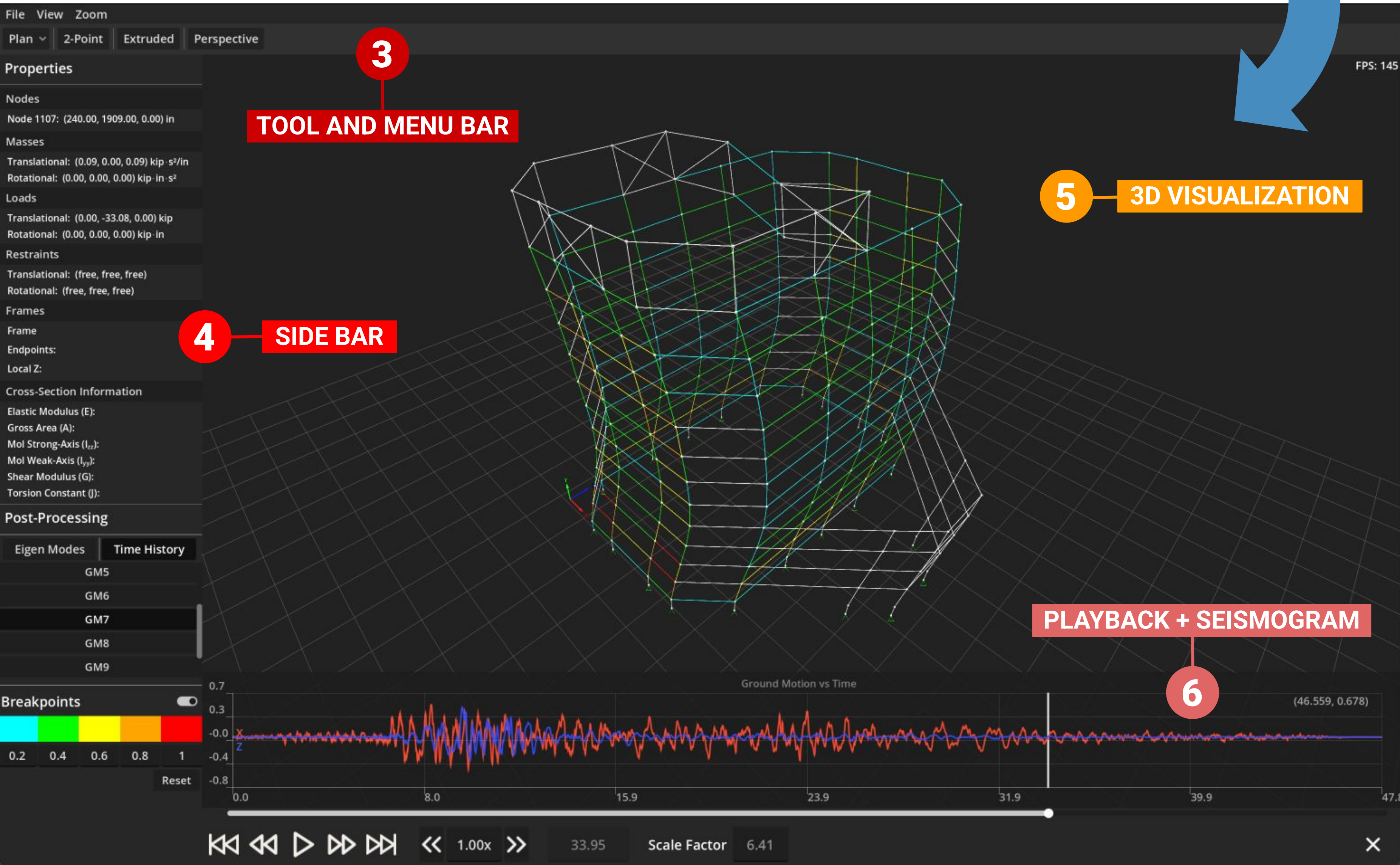
OPENSEES INPUT SCRIPT

1

```
20 0.2 6.53302e-07 -1.28862e-06 1.44272e-06 6.53302e-07 6.53303e-07 -5.27
21 0.21 6.44314e-07 -1.28871e-06 1.44391e-06 6.44314e-07 6.44314e-07 -5.2
22 0.22 6.37059e-07 -1.28866e-06 1.44596e-06 6.37059e-07 6.3706e-07 -5.25
23 0.23 6.32574e-07 -1.28805e-06 1.44852e-06 6.32574e-07 6.32574e-07 -5.2
24 0.24 6.31713e-07 -1.28643e-06 1.4512e-06 6.31713e-07 6.31714e-07 -5.23
25 0.25 6.35042e-07 -1.28643e-06 1.4512e-06 6.35043e-07 6.35043e-07 -5.2
26 0.26 6.42762e-07 -1.27908e-06 1.45595e-06 6.42762e-07 6.42763e-07 -5.2
27 0.27 6.54688e-07 -1.2733e-06 1.458e-06 6.54688e-07 6.54688e-07 -5.2096
28 0.28 6.70277e-07 -1.26632e-06 1.46014e-06 6.70277e-07 6.70277e-07 -5.1
29 0.29 6.88708e-07 -1.25839e-06 1.46279e-06 6.88708e-07 6.88709e-07 -5.1
30 0.3 7.09e-07 -1.24971e-06 1.46638e-06 7.09e-07 7.09001e-07 -5.17334e-6
31 0.31 7.30153e-07 -1.24041e-06 1.4713e-06 7.30153e-07 7.30153e-07 -5.15
32 0.32 7.51291e-07 -1.23054e-06 1.47782e-06 7.51291e-07 7.51292e-07 -5.1
33 0.33 7.71794e-07 -1.22015e-06 1.48609e-06 7.71794e-07 7.71795e-07 -5.1
34 0.34 7.91365e-07 -1.2093e-06 1.49608e-06 7.91365e-07 7.91365e-07 -5.09
```

OPENSEES OUTPUT

2



3

TOOL AND MENU BAR

4

SIDE BAR

5

3D VISUALIZATION

PLAYBACK + SEISMOGRAM

6

FIGURE KEY

Supplied by KPFF:

1. Script defining model properties.
2. Raw analysis output data including node displacements, forces and response values.

Built by our team:

3. Controls for **file import**, camera **views** and settings, and adjustable **zoom** settings.
4. Toggable **properties** for elements, masses, loads, restraints, and cross-section. Real-time **DCR** color scaling and adjustable **breakpoints**. **Eigen modes** and **time history** analyses.
5. Interactive 3D model viewport with real-time **pan**, **rotate**, **zoom** and **inspection** for pre- and post-processing.
6. Playback controls (**time-step**, **speed**, **progress**) with adjustable x/z Ground Motion vs Time plot.

FUTURE WORK

- Add **group** and **elevation** view modes.
- **OpenSees backend integration**.
- 3D cityscape overlay for client context.
- Display **node** and **beam tags** on the model.
- Show **directional load arrows** for clarity.
- Add plastic hinges, dampers, and nonlinear wall elements, as well as build tools for pushover analysis and **performance checking**.

REFERENCES & ACKNOWLEDGEMENTS

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- We gratefully acknowledge the open-source software tools used in our development: OpenSees, Godot Engine, LD2Studio - graph2D.

QUESTIONS?
CONTACT US!

